CSA4704-DEEP LEARNING

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Experiment 21: Factor Analysis and Latent Factor Interpretation

Problem Statement: An HR department runs a 10-question employee-satisfaction survey. Management wants to know which few hidden factors drive the answers. Survey responses from 300 employees on pay, workload, manager support, growth, team spirit and similar items (Likert 1–5). The objective is to identify and interpret latent factors with factor analysis. Use Scikit-learn, NumPy, Pandas, Matplotlib.

Tasks:

1. Load and explore the survey data in Pandas; check correlations.

2. Fit Scikit-learn's FactorAnalysis with 2, 3 and 4 factors.

3. Apply varimax rotation and display the loadings matrix as a heatmap.

4. Name each factor from its high-loading questions (for example, "Compensation", "Work Culture").

5. Compare factor analysis with PCA on the same data.

FORMULA:

1. Mean Calculation

Mean = Sum of all values / Number of observations

2. Correlation Matrix

$r = Cov(X,Y) / (SD(X) \times SD(Y))$

3. Factor Analysis Model

$X = LF + E$

4. Factor Loading Matrix

$X = L \times F + E$

5. Covariance Matrix

$\Sigma = L \times L.T + \Psi$

6. Factor Score

Factor Score = X × Factor Loadings

7. Varimax Rotation

Maximize V = Sum of [Variance of squared factor loadings]

8. Communality

h^2 = Sum of (Factor Loadings^2)

9. Uniqueness

Uniqueness = 1 - Communality

10. PCA Explained Variance Ratio

Explained Variance Ratio = Eigenvalue / Sum of All Eigenvalues

11. Cumulative Explained Variance

Cumulative Variance = Sum of Explained Variance Ratios

CODE:

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.preprocessing import StandardScaler

from sklearn.decomposition import FactorAnalysis, PCA

np.random.seed(42)

# 1. Generate sample survey data for 300 employees

n = 300

questions = [

    "Salary Satisfaction",

    "Benefits Satisfaction",

    "Workload Balance",

    "Work Pressure",

    "Manager Support",

    "Communication",

    "Career Growth",

    "Learning Opportunities",

    "Team Cooperation",
```

```
    "Workplace Belonging"
]
# Three hidden factors: Compensation, Workload, and Work Culture
latent = np.random.randn(n, 3)
loadings_true = np.array([
    [0.85, 0.10, 0.10],
    [0.80, 0.10, 0.15],
    [0.10, -0.85, 0.10],
    [0.10, -0.80, 0.10],
    [0.15, 0.10, 0.85],
    [0.15, 0.10, 0.80],
    [0.20, 0.15, 0.75],
    [0.20, 0.15, 0.70],
    [0.10, 0.15, 0.85],
    [0.10, 0.15, 0.80]
])
continuous_data = latent @ loadings_true.T
continuous_data += np.random.normal(0, 0.5, (n, 10))
# Convert values to Likert scale 1–5
survey = np.clip(
```

```
    np.rint(3 + continuous_data), 1, 5
).astype(int)
df = pd.DataFrame(survey, columns=questions)
print("First 5 survey responses:")
print(df.head())
print("\nDataset information:")
print(df.info())
print("\nSummary statistics:")
print(df.describe().round(2))
# 2. Check correlations between survey questions
correlation = df.corr()
plt.figure(figsize=(11, 8))
sns.heatmap(correlation, annot=True, cmap="coolwarm",
            vmin=-1, vmax=1, fmt=".2f")
plt.title("Correlation Between Survey Questions")
plt.tight_layout()
plt.show()
# Standardize the survey data
scaler = StandardScaler()
X = scaler.fit_transform(df)
```

```
# 3. Varimax rotation function
def varimax(Phi, gamma=1.0, q=100, tol=1e-6):
    p, k = Phi.shape
    R = np.eye(k)
    old_objective = 0
    for _ in range(q):
        loadings = Phi @ R
        u, s, vh = np.linalg.svd(
            Phi.T @ (
                loadings**3
                - (gamma / p) * loadings
                @ np.diag(np.diag(loadings.T @ loadings))
            )
        )
        R = u @ vh
        objective = np.sum(s)
        if old_objective != 0 and (
            objective - old_objective < tol * old_objective
        ):
            break
```

```
        old_objective = objective
    return Phi @ R
# 4. Fit Factor Analysis with 2, 3 and 4 factors
factor_models = {}
rotated_loadings = {}
for k in [2, 3, 4]:
    model = FactorAnalysis(n_components=k, random_state=42)
   model.fit(X)
    # sklearn components_: factors x features
    loadings = model.components_.T
    # Apply varimax rotation
    rotated = varimax(loadings)
    factor_models[k] = model
    rotated_loadings[k] = rotated
    print(f"\nFactor Analysis with {k} factors")
    print("Log-likelihood:", round(model.score(X), 4))
    loading_df = pd.DataFrame(
        rotated,
        index=questions,
```

```
        columns=[f"Factor {i+1}" for i in range(k)]
    )
    print(loading_df.round(3))
    # Display loadings heatmap
    plt.figure(figsize=(8, 6))
    sns.heatmap(loading_df, annot=True, cmap="coolwarm",
                center=0, fmt=".2f")
    plt.title(f"Varimax-Rotated Factor Loadings ({k} Factors)")
    plt.tight_layout()
    plt.show()

# 5. Interpret the three-factor model
loadings_3 = pd.DataFrame(
    rotated_loadings[3],
    index=questions,
    columns=["Factor 1", "Factor 2", "Factor 3"]
)
print("\nFACTOR INTERPRETATION (3-factor model)")
for factor in loadings_3.columns:
    strongest = loadings_3[factor].abs().sort_values(
```

```
        ascending=False
    ).head(4)
    print(f"\n{factor} - highest-loading questions:")
    for question in strongest.index:
        print(f"  {question}: {loadings_3.loc[question, factor]:.3f}")
# 6. Fit PCA on the same standardized data
pca = PCA(n_components=4)
pca.fit(X)


pca_loadings = pca.components_.T * np.sqrt(
    pca.explained_variance_
)
pca_df = pd.DataFrame(
    pca_loadings,
    index=questions,
    columns=[f"PC{i+1}" for i in range(4)]
)
print("\nPCA loadings:")
print(pca_df.round(3))
print("\nPCA explained variance ratio:")
```

```
for i, ratio in enumerate(pca.explained_variance_ratio_, 1):
    print(f"PC{i}: {ratio:.3f}")
plt.figure(figsize=(8, 6))
sns.heatmap(pca_df, annot=True, cmap="coolwarm",
            center=0, fmt=".2f")
plt.title("PCA Loadings")
plt.tight_layout()
plt.show()
# 7. Compare Factor Analysis and PCA
print("\nCOMPARISON")
print("Factor Analysis models common latent factors and estimates")
print("specific/residual variance for each question.")
print("PCA finds directions that explain total data variance.")
plt.figure(figsize=(8, 5))
plt.plot(
    range(1, 5),
    np.cumsum(pca.explained_variance_ratio_),
    marker="o"
)
plt.xlabel("Number of Principal Components")
```

```
plt.ylabel("Cumulative Explained Variance")

plt.title("PCA Explained Variance")

plt.xticks(range(1, 5))

plt.grid(True)

plt.show()

print("\nExperiment 21 completed successfully.")
```

OUTPUTl:

First 5 survey responses:

	Salary Satisfaction	Benefits Satisfaction	Workload Balance
0	4	3	3
1	4	4	3
2	5	4	3
3	3	4	3
4	2	3	5

	Work Pressure	Manager Support	Communication	Career Growth
0	4	4	4	3
1	4	3	3	3
2	2	4	2	3
3	2	3	2	2

4 4 2 1 1

	Learning Opportunities	Team Cooperation	Workplace Belonging
0	4	4	5
1	3	3	3
2	3	4	3
3	3	3	2
4	1	1	1

Dataset information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 300 entries, 0 to 299

Data columns (total 10 columns):

#	Column	Non-Null Count	Dtype
0	Salary Satisfaction	300 non-null	int64
1	Benefits Satisfaction	300 non-null	int64
2	Workload Balance	300 non-null	int64
3	Work Pressure	300 non-null	int64
4	Manager Support	300 non-null	int64
5	Communication	300 non-null	int64
6	Career Growth	300 non-null	int64

7 Learning Opportunities 300 non-null int64

8 Team Cooperation 300 non-null int64

9 Workplace Belonging 300 non-null int64

dtypes: int64(10)

memory usage: 23.6 KB

None

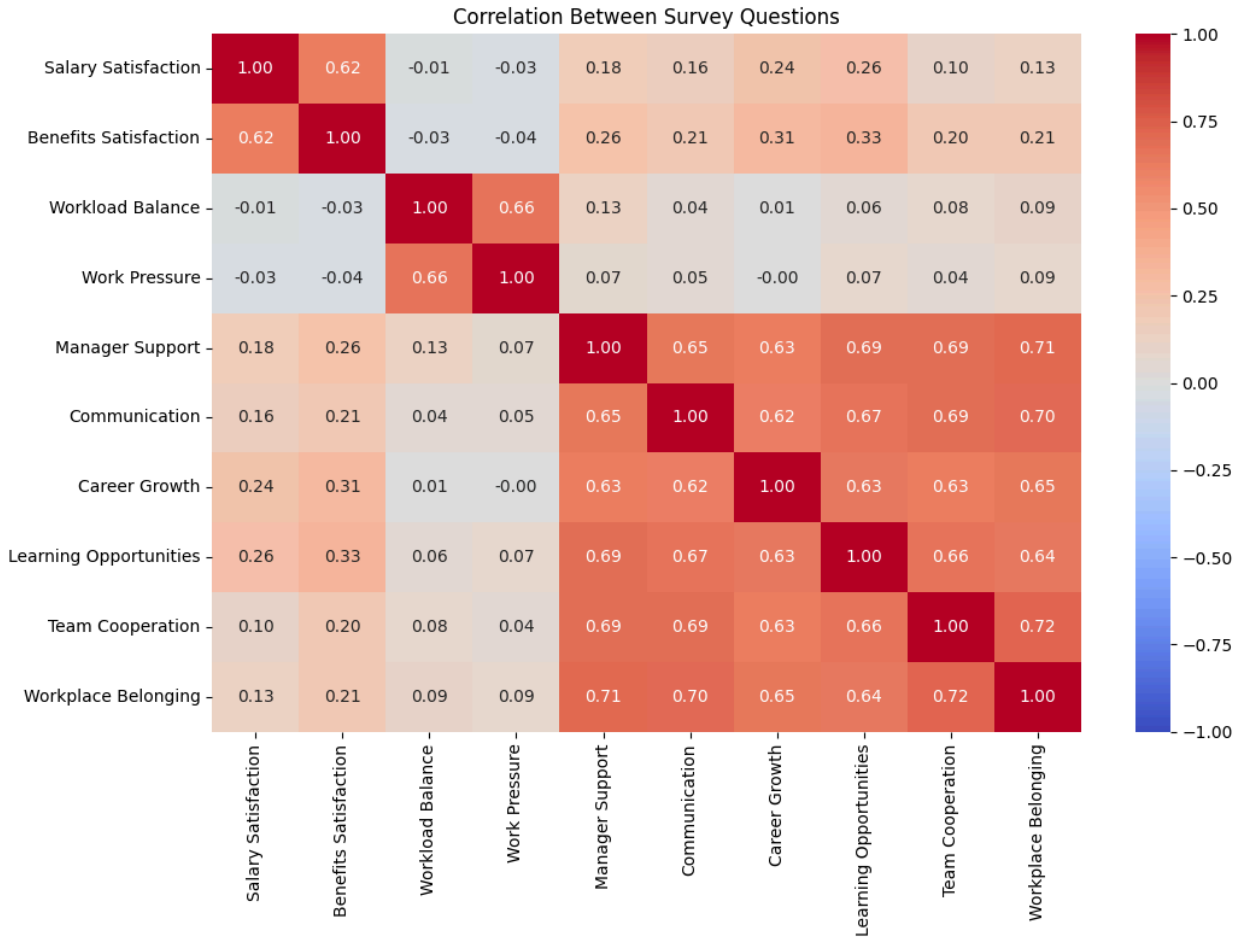
Summary statistics:

	Salary Satisfaction	Benefits Satisfaction	Workload Balance \
count	300.00	300.00	300.00
mean	3.08	3.05	3.06
std	0.92	0.94	0.97
min	1.00	1.00	1.00
25%	2.75	2.00	2.75
50%	3.00	3.00	3.00
75%	4.00	4.00	4.00
max	5.00	5.00	5.00

	Work Pressure	Manager Support	Communication	Career Growth \
count	300.00	300.00	300.00	300.00
mean	3.05	3.05	3.03	3.01
std	0.99	1.02	1.02	0.97

min	1.00	1.00	1.00	1.00
25%	2.00	2.00	2.00	2.00
50%	3.00	3.00	3.00	3.00
75%	4.00	4.00	4.00	4.00
max	5.00	5.00	5.00	5.00

	Learning Opportunities	Team Cooperation	Workplace Belonging
count	300.00	300.00	300.00
mean	3.04	3.07	3.12
std	0.93	1.01	0.97
min	1.00	1.00	1.00
25%	2.00	2.00	2.75
50%	3.00	3.00	3.00
75%	4.00	4.00	4.00
max	5.00	5.00	5.00

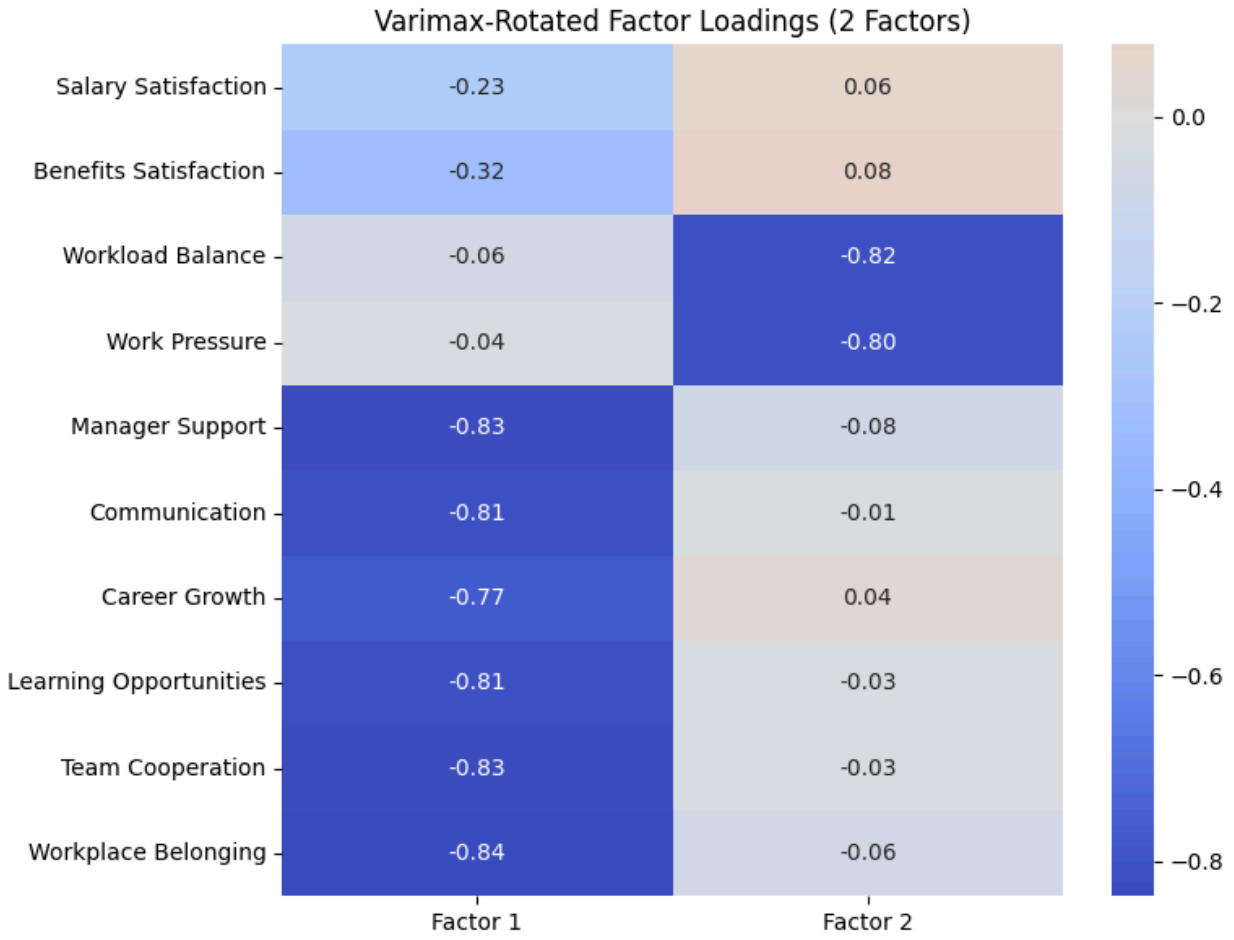


Factor Analysis with 2 factors

Log-likelihood: -11.8054

	Factor 1	Factor 2
Salary Satisfaction	-0.230	0.058
Benefits Satisfaction	-0.320	0.078
Workload Balance	-0.055	-0.817
Work Pressure	-0.036	-0.803
Manager Support	-0.826	-0.077
Communication	-0.813	-0.009

Career Growth	-0.772	0.045
Learning Opportunities	-0.807	-0.030
Team Cooperation	-0.831	-0.035
Workplace Belonging	-0.838	-0.065

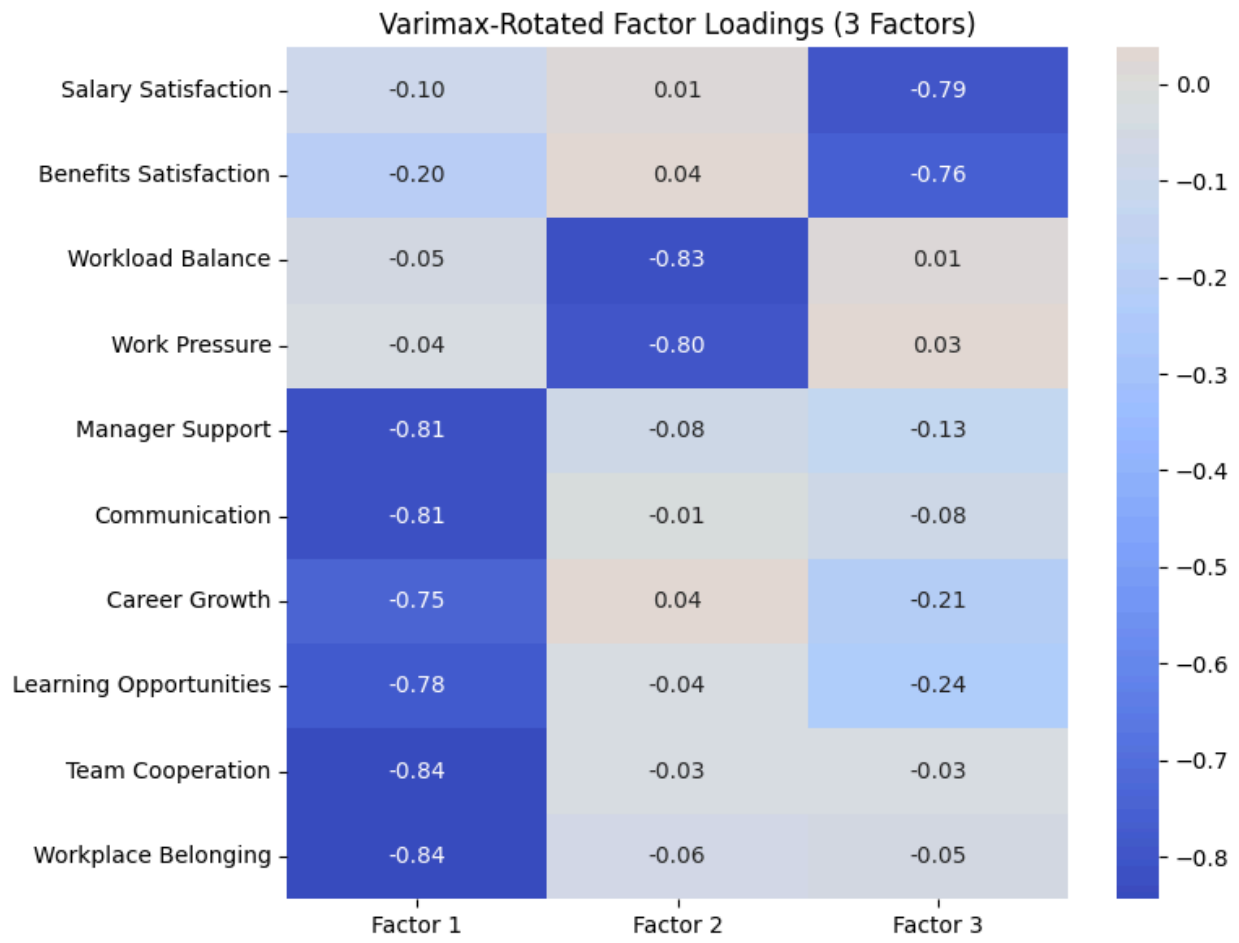


Factor Analysis with 3 factors

Log-likelihood: -11.5423

	Factor 1	Factor 2	Factor 3
Salary Satisfaction	-0.099	0.011	-0.794

Benefits Satisfaction	-0.201	0.038	-0.758
Workload Balance	-0.054	-0.826	0.012
Work Pressure	-0.037	-0.796	0.029
Manager Support	-0.814	-0.082	-0.131
Communication	-0.811	-0.008	-0.082
Career Growth	-0.745	0.035	-0.212
Learning Opportunities	-0.777	-0.041	-0.237
Team Cooperation	-0.843	-0.031	-0.030
Workplace Belonging	-0.844	-0.062	-0.053

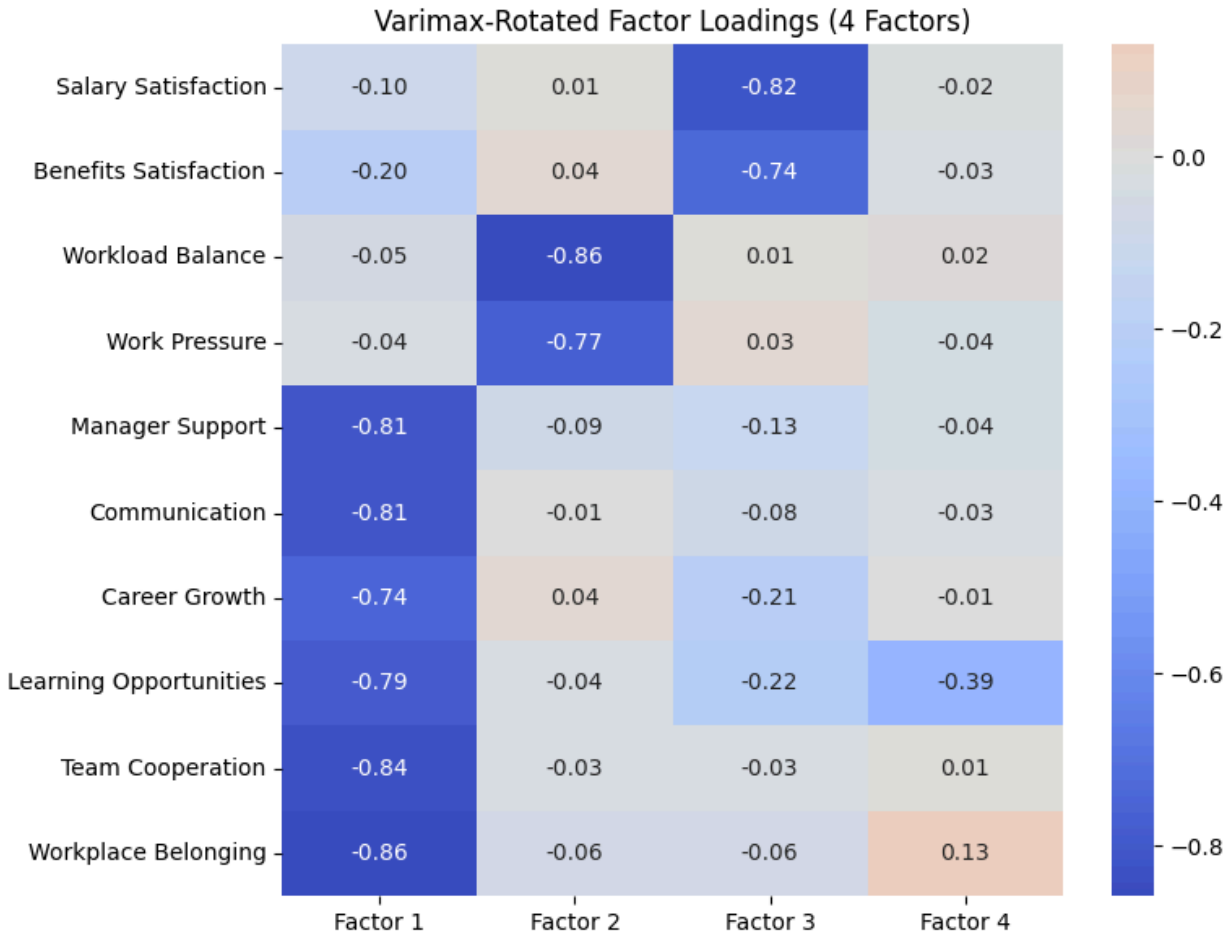


Factor Analysis with 4 factors

Log-likelihood: -11.5287

	Factor 1	Factor 2	Factor 3	Factor 4
Salary Satisfaction	-0.098	0.010	-0.816	-0.017
Benefits Satisfaction	-0.204	0.039	-0.738	-0.033
Workload Balance	-0.052	-0.858	0.009	0.022
Work Pressure	-0.039	-0.767	0.034	-0.040
Manager Support	-0.810	-0.086	-0.128	-0.040
Communication	-0.807	-0.005	-0.081	-0.030

Career Growth	-0.744	0.035	-0.210	-0.007
Learning Opportunities	-0.786	-0.037	-0.218	-0.385
Team Cooperation	-0.839	-0.034	-0.030	0.010
Workplace Belonging	-0.858	-0.061	-0.057	0.130



FACTOR INTERPRETATION (3-factor model)

Factor 1 - highest-loading questions:

Workplace Belonging: -0.844

Team Cooperation: -0.843

Manager Support: -0.814

Communication: -0.811

Factor 2 - highest-loading questions:

Workload Balance: -0.826

Work Pressure: -0.796

Manager Support: -0.082

Workplace Belonging: -0.062

Factor 3 - highest-loading questions:

Salary Satisfaction: -0.794

Benefits Satisfaction: -0.758

Learning Opportunities: -0.237

Career Growth: -0.212

PCA loadings:

	PC1	PC2	PC3	PC4
Salary Satisfaction	0.327	-0.383	0.754	0.051
Benefits Satisfaction	0.418	-0.392	0.688	-0.119
Workload Balance	0.109	0.831	0.363	-0.052
Work Pressure	0.088	0.836	0.351	0.102

Manager Support	0.852	0.071	-0.094	-0.183
Communication	0.834	0.024	-0.176	-0.062
Career Growth	0.815	-0.090	-0.052	0.543
Learning Opportunities	0.849	-0.017	0.000	-0.121
Team Cooperation	0.843	0.068	-0.217	-0.126
Workplace Belonging	0.852	0.089	-0.179	0.005

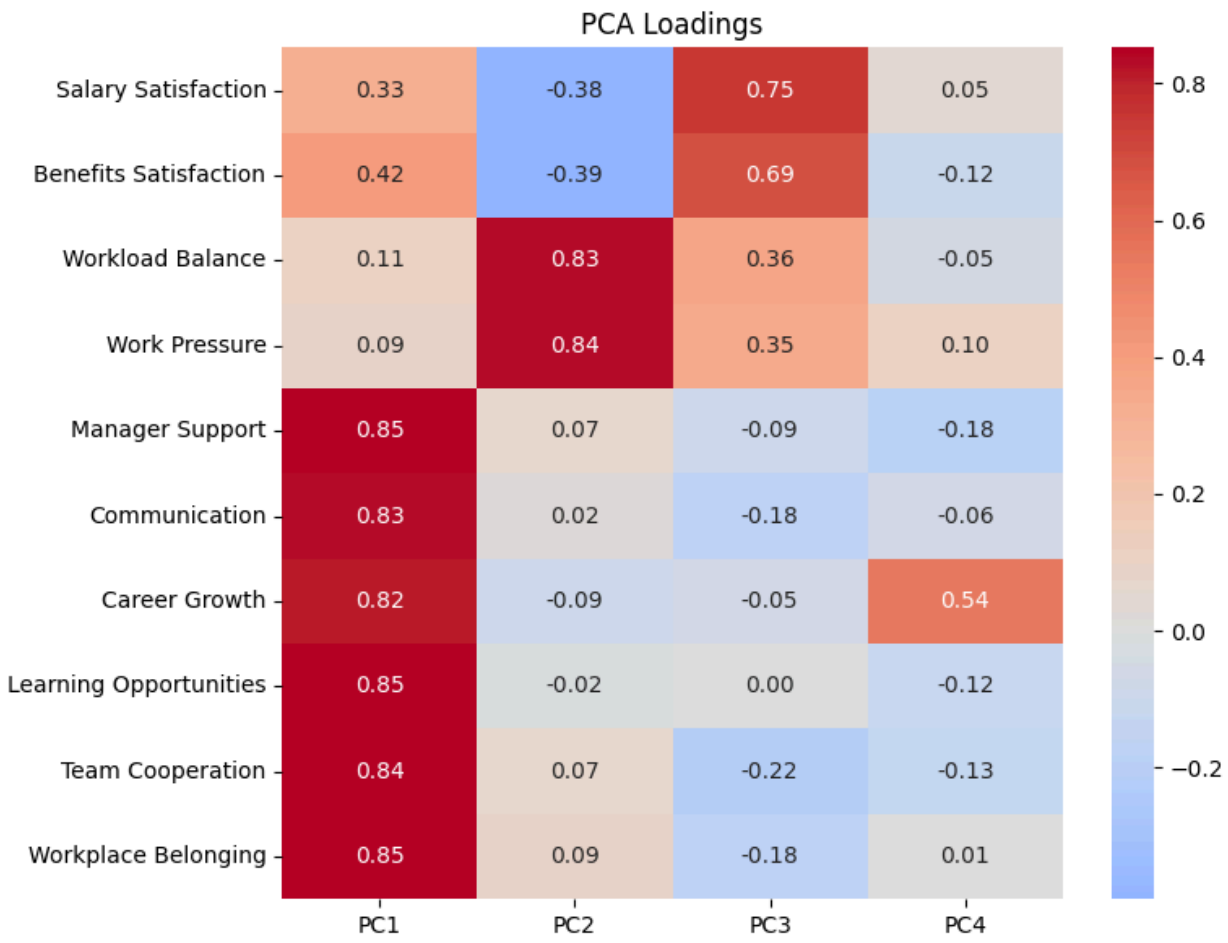
PCA explained variance ratio:

PC1: 0.453

PC2: 0.171

PC3: 0.141

PC4: 0.039

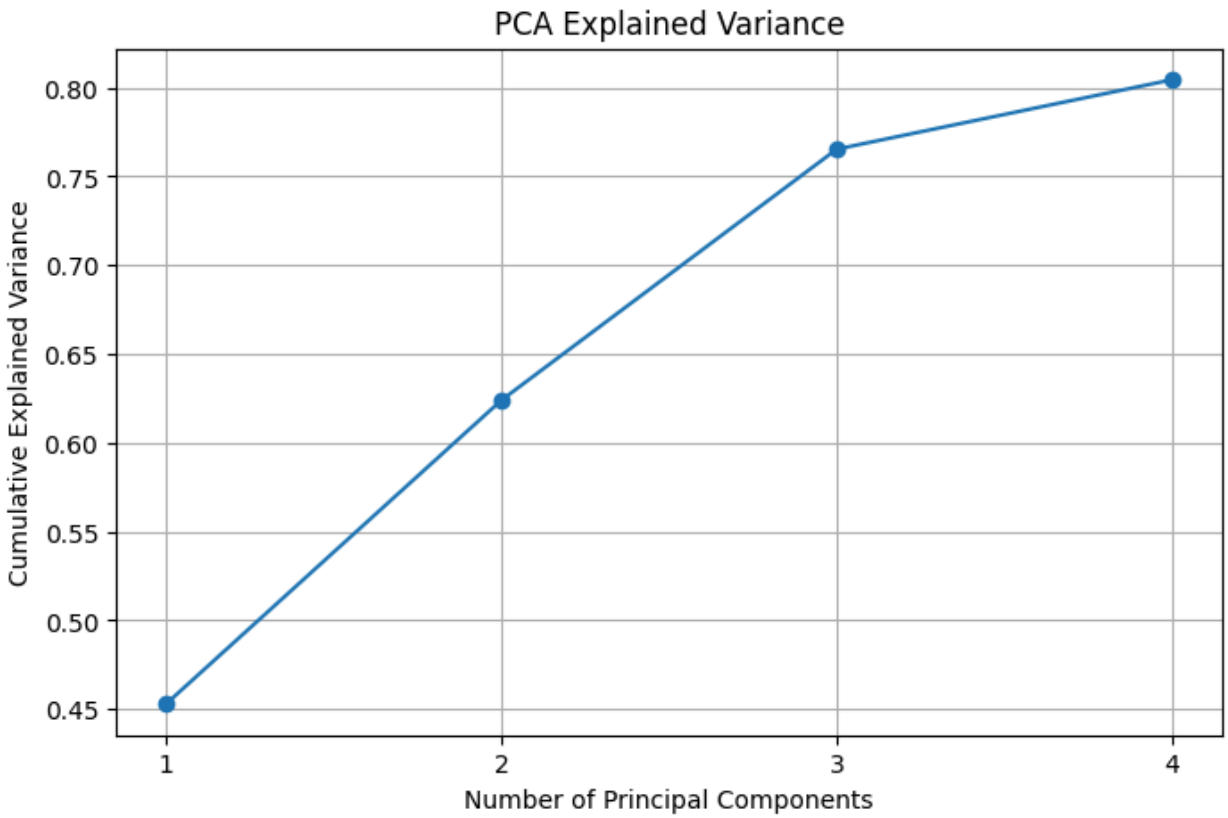


COMPARISON

Factor Analysis models common latent factors and estimates

specific/residual variance for each question.

PCA finds directions that explain total data variance.


PCA Explained Variance
Cumulative Explained Variance
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.45
1
2
3
4
Number of Principal Components